

Machine Learning

Unit-I (Part-B): Linear Regression

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Linear Regression

- Linear regression is a **type of supervised machine-learning algorithm** that learns from the labelled datasets.
- It maps the data points with most optimized linear functions which can be **used for prediction on new datasets**.
- It assumes that there is a **linear relationship between the input and output**, meaning the **output changes at a constant rate as the input changes**.
- This relationship is **represented by a straight line**.

Linear Regression

Example:

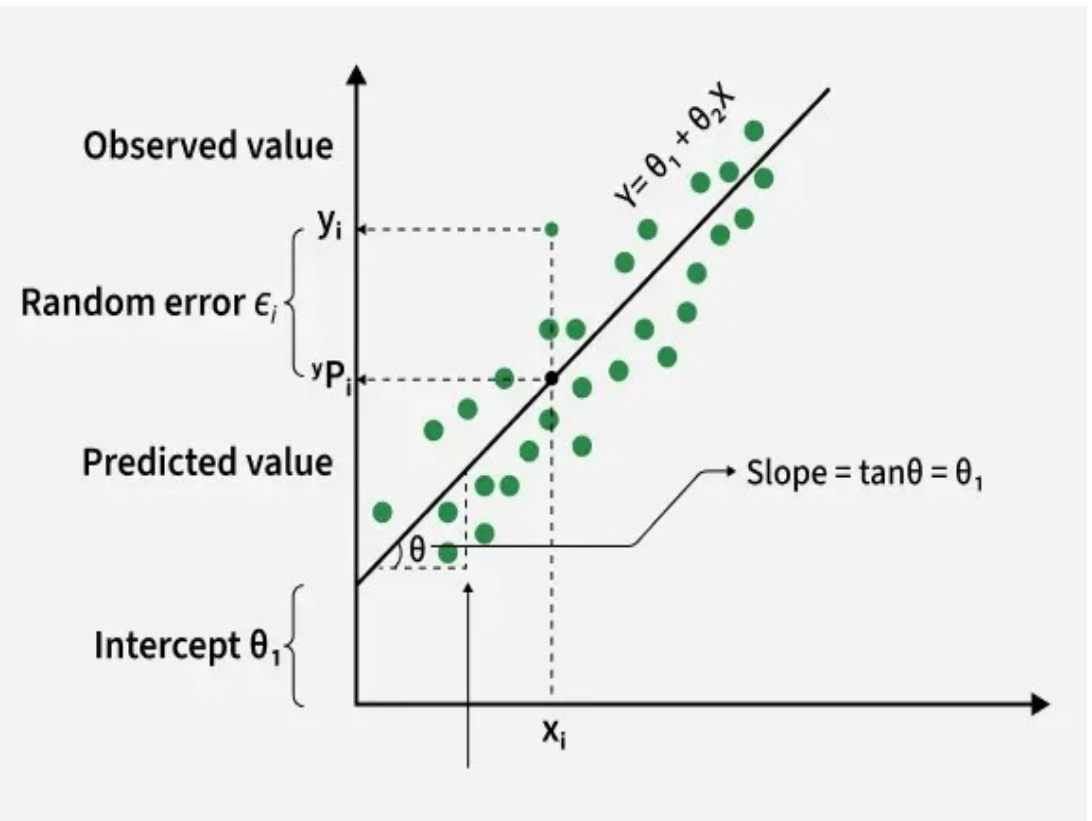
- We want to **predict a student's exam score based on how many hours they studied.**
We observe that as students study more hours, their scores go up. In the example of predicting exam scores based on hours studied.
- Here,
 - **Independent variable (input):** **Hours** studied because it's the factor we control or observe.
 - **Dependent variable (output):** **Exam score** because it depends on how many hours were studied.
 - We **use the independent variable to predict the dependent variable.**

Best Fit Line in Linear Regression

- The best-fit line is the straight line that most accurately represents the relationship between the independent variable (input) and the dependent variable (output).
- It is the line that minimizes the difference between the actual data points and the predicted values from the model.

Goal of the Best-Fit Line

- The goal of linear regression is to find a straight line that minimizes the error (the difference) between the observed data points and the predicted values.
- This line helps us predict the dependent variable for new, unseen data.
- Here, Y is called a dependent or target variable and X is called an independent variable also known as the predictor of Y .
- There are many types of functions or modules that can be used for regression.
- A linear function is the simplest type of function.
- Here, X may be a single feature or multiple features representing the problem.



Equation of the Best-Fit Line

- For simple linear regression (with one independent variable), the best-fit line is represented by the equation

$$y = b + mx$$

- Where:
 - y is the predicted value (dependent variable)
 - x is the input (independent variable)
 - m is the slope of the line (how much y changes when x changes)
 - m is also called regression coefficient.
 - b is the intercept (the value of y when $x = 0$)
- The best-fit line will be the one that optimizes the values of m (slope) and b (intercept) so that the predicted y values are as close as possible to the actual data points.

Equation of the Best-Fit Line

- Intercept value, b , and slope of the line, m , are evaluated using the formulas given below:

$$b = \frac{\sum y \sum x^2 - \sum x \sum xy}{n(\sum x^2) - (\sum x)^2}$$

$$m = \frac{n \sum xy - (\sum x)(\sum y)}{n(\sum x^2) - (\sum x)^2}$$

- These formulas are derived from minimizing the sum of squared errors:

$$Error = \sum (y_i - (mx_i + b))^2$$

- The values of m and b that minimize this error give the best-fitting straight line.
- $\sum xy$ represents correlation between x and y .
- $\sum x^2$ represents variance of x .

Minimizing the Error: The Least Squares Method

- To find the best-fit line, we use a method called **Least Squares**.
- The idea behind this method is to **minimize the sum of squared differences between the actual values (data points) and the predicted values from the line**.
- These differences are called **residuals**.
- The formula for residuals is:

$$\text{Residual} = y_i - \hat{y}_i$$

- Where:
 - y_i is the actual observed value
 - $\hat{y}_i$ is the predicted value from the line for that x_i

Minimizing the Error: The Least Squares Method

- The least squares method **minimizes the sum of the squared residuals**:

$$\text{Sum of squared errors (SSE)} = \sum (y_i - \hat{y}_i)^2$$

- This method ensures that the **line best represents the data** where the sum of the squared differences between the predicted values and actual values is **as small as possible**.

Interpretation of the Best-Fit Line

Slope (m):

- The slope of the best-fit line indicates how much the dependent variable (y) changes with each unit change in the independent variable (x).
- For example if the slope is 5, it means that for every 1-unit increase in x, the value of y increases by 5 units.

Intercept (b):

- The intercept represents the predicted value of y when $x = 0$. It's the point where the line crosses the y-axis.

Example 1

Find the linear regression equation for the given data:

x	y
3	8
9	6
5	4
3	2

Solution 1

x	y	x²	xy
3	8	9	24
9	6	81	54
5	4	25	20
3	2	9	6
$\sum x = 20$	$\sum y = 20$	$\sum x^2 = 124$	$\sum xy = 104$

Solution 1

1. Calculating intercept and slope value.

2. Using formula,

$$3. \quad b = \frac{\sum y \sum x^2 - \sum x \sum xy}{n(\sum x^2) - (\sum x)^2}$$

$$4. \quad b = \frac{\{20(124) - 20(104)\}}{\{4(124) - 400\}}$$

$$5. \quad b = \frac{400}{96} = 4.17$$

$$6. \quad m = \frac{n \sum xy - (\sum x)(\sum y)}{n \sum x^2 - (\sum x)^2}$$

$$7. \quad m = \frac{\{4(104) - 20(20)\}}{\{4(124) - 400\}}$$

$$8. \quad m = \frac{16}{96} = 0.166$$

9. So, linear regression equation is,

$$10. \quad y = b + mx \rightarrow y = 4.17 + 0.167x$$

Question 1

Find the linear regression equation for the given data:

x	y
4	6
7	5
3	8
1	3

Solution 1

$$y = 4.8 + 0.187x$$

Question 2

Find the intercept of the linear regression line if $\sum x = 25$, $\sum y = 20$, $\sum x^2 = 90$, $\sum xy = 150$, and $n = 5$.

Solution 2

$$b = 11.14$$

Question 3

Find slope of linear regression line if $\sum x = 10$, $\sum y = 16$, $\sum x^2 = 60$, $\sum xy = 120$ and $n = 4$.

Solution 3

$$m = 2.29$$

Question 4

Predict the house price based on house size of 1600 square feet for the following data:

Size (square feet)	Price (₹ Lakhs)
800	40
1000	50
1200	65
1500	80
1800	95

Solution 4

- Linear Regression Equation:

$$Price = (-4.177) + 0.055 * Size$$

- predicted price $\approx$ ₹83.823 Lakhs

Question 5

A company's stock price depends on the number of days after its IPO. The following table shows the stock price for the first few days after listing:

Day (X)	Stock Price (Y in ₹)
1	120
2	125
3	130
4	140
5	145

Question 5

1. Find the regression equation using Simple Linear Regression.
2. Using the obtained regression equation, predict the stock price for the next 5 days (Day 6 to Day 10).
3. Update the table by removing the existing stock prices and by including the predicted stock prices; and again calculate the regression equation.
4. Compare both regression equations and comment on whether they are similar or not.

Solution 5

1. $Y = 112.5 + 6.5X$

Day (X)	Predicted Price (Y)
6	151.5
7	158.0
8	164.5
9	171.0
10	177.5

4. New Regression Equation (using New predicted values):

$$Y = 112.5 + 6.5X$$

Limitations

- **Assumes Linearity:** The method assumes the relationship between the variables is linear. If the relationship is non-linear, linear regression might not work well.
- **Sensitivity to Outliers:** Outliers can significantly affect the slope and intercept, skewing the best-fit line.

Hypothesis Function in Linear Regression

- In linear regression, the hypothesis function is the equation used to make predictions about the dependent variable based on the independent variables.
- It represents the relationship between the input features and the target output.
- For a simple case with one independent variable, the hypothesis function is:

$$h(x) = \beta_0 + \beta_1 x$$

- Where:
 - $h(x)$ or $(\hat{y})$ is the predicted value of the dependent variable (y).
 - x is the independent variable.
 - β_0 is the intercept, representing the value of y when x is 0.
 - β_1 is the slope, indicating how much y changes for each unit change in x .

Hypothesis Function in Linear Regression

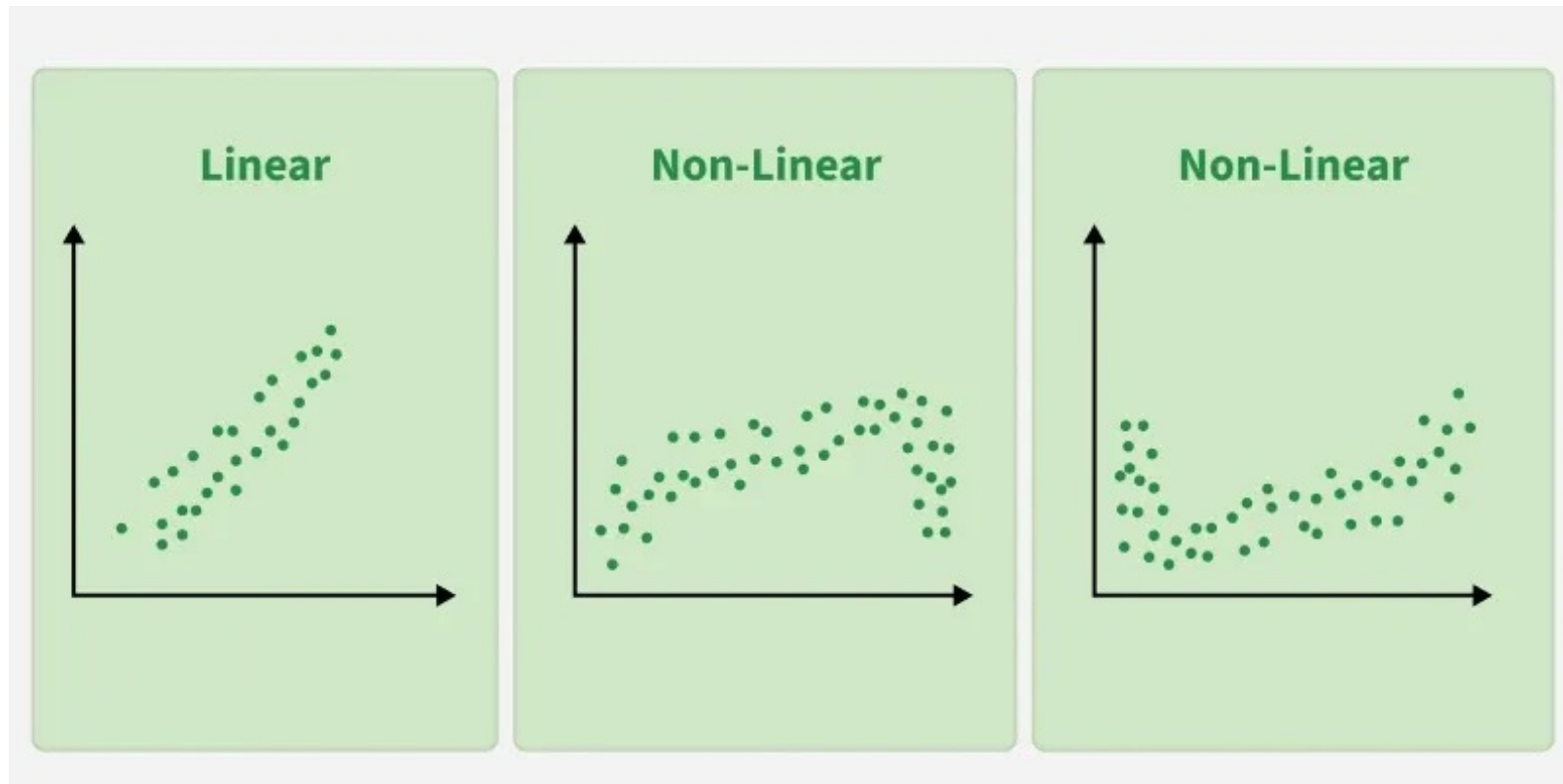
- For multiple linear regression (with more than one independent variable), the hypothesis function expands to:

$$h(x_1, x_2, \dots, x_k) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k$$

- Where:
 - $x_1, x_2, \dots, x_k$ are the independent variables.
 - β_0 is the intercept.
 - $\beta_1, \beta_2, \dots, \beta_k$ are the coefficients, representing the influence of each respective independent variable on the predicted output.

Assumptions of the Linear Regression

1. Linearity: The relationship between inputs (X) and the output (Y) is a straight line.



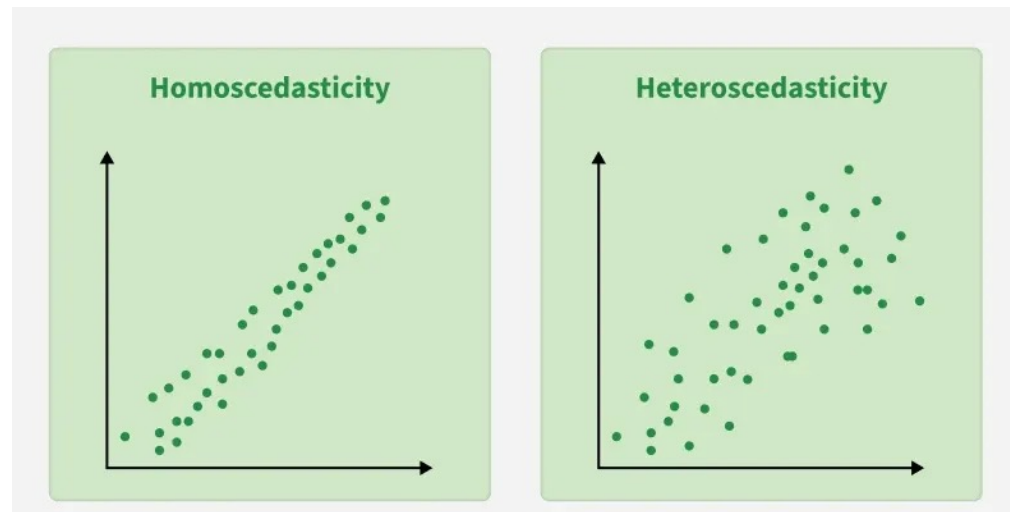
Assumptions of the Linear Regression

2. Independence of Errors:

- The prediction **errors should be independent of each other**. For example, if your model makes a wrong prediction for one data point, it should not affect the prediction for the next data point.

3. Constant Variance (Homoscedasticity):

- The errors should have **equal spread across all values of the input**.
- If the spread changes (like fans out or shrinks), it's called heteroscedasticity and it's a problem for the model.



Assumptions of the Linear Regression

4. Normality of Errors:

- The prediction errors (residuals) should follow a normal (bell-shaped) distribution.

5. No Multicollinearity(for multiple regression):

- Input variables shouldn't be too closely related to each other.

6. No Autocorrelation:

- Errors shouldn't show repeating patterns, especially in time-based data.
- For example, An error today should not depend on yesterday's error (Stock Price).

7. Additivity:

- The total effect on Y is just the sum of effects from each X, no mixing or interaction between them.

Types of Linear Regression

- When there is **only one independent feature** it is known as Simple Linear Regression or **Univariate Linear Regression** and,
- When there are **more than one independent feature** it is known as Multiple Linear Regression or **Multivariate Regression**.

Simple Linear Regression

- Simple linear regression is used when we want to predict a target value (dependent variable) **using only one input feature** (independent variable).
- It assumes a straight-line relationship between the two.
- **Formula:** $\hat{y} = \theta_0 + \theta_1 x$
- **Example:** Predicting a person's salary (y) based on their years of experience (x).

Multiple Linear Regression

- Multiple linear regression involves more than one independent variable and one dependent variable.
- The equation for multiple linear regression is:

$$\hat{y} = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_n x_n$$

- The goal is to find the best-fit line that predicts Y accurately for given inputs X.

Example

Suppose we want to predict a student's **Final Score (Y)** based on:

- X_1 = Hours Studied
- X_2 = Attendance (%)

Find the intercept and regression coefficients for input variables.

Student	Hours Studied (X_1)	Attendance (X_2)	Final Score (Y)
1	2	60	65
2	4	70	75
3	6	80	85
4	8	90	95

Step-1: Regression Model

$$Y = b_0 + b_1X_1 + b_2X_2$$

Where:

- $b_0 \rightarrow$ Intercept
- $b_1 \rightarrow$ Coefficient of Hours Studied
- $b_2 \rightarrow$ Coefficient of Attendance

Step-2: Construct Normal Equations

For multiple linear regression, coefficients can be found using **normal equations**:

Equation 1 (Overall balance): This equation ensures that the **total predicted values match the total actual values**.

$$\sum Y = nb_0 + b_1 \sum X_1 + b_2 \sum X_2$$

Equation 2 (Relation with X_1): This **connects Hours Studied with Final Score**.

$$\sum X_1 Y = b_0 \sum X_1 + b_1 \sum X_1^2 + b_2 \sum X_1 X_2$$

Equation 3 (Relation with X_2): This **connects Attendance with Final Score**.

$$\sum X_2 Y = b_0 \sum X_2 + b_1 \sum X_1 X_2 + b_2 \sum X_2^2$$

Step-3: Create Calculation Table

X_1	X_2	Y	X_1^2	X_2^2	X_1X_2	X_1Y	X_2Y
2	60	65	4	3600	120	130	3900
4	70	75	16	4900	280	300	5250
6	80	85	36	6400	480	510	6800
8	90	95	64	8100	720	760	8550
$\Sigma = 20$	$\Sigma = 300$	$\Sigma = 320$	$\Sigma = 120$	$\Sigma = 23000$	$\Sigma = 1600$	$\Sigma = 1700$	$\Sigma = 24500$

Step-4: Substitute into Normal Equations

- $320 = 4b_0 + 20b_1 + 300b_2$
- $1700 = 20b_0 + 120b_1 + 1600b_2$
- $24500 = 300b_0 + 1600b_1 + 23000b_2$

Step-5: Solve the Equations:

(using elimination method)

Equation (1):

$$4b_0 + 20b_1 + 300b_2 = 320$$

Equation (2):

$$20b_0 + 120b_1 + 1600b_2 = 1700$$

Equation (3):

$$300b_0 + 1600b_1 + 23000b_2 = 24500$$

Solving step-by-step gives:

$$b_0 = 30, b_1 = 2.5, b_2 = 0.5$$

Final Regression Equation

$$Y = 30 + 2.5X_1 + 0.5X_2$$

Question 1

A company wants to predict **Profit (Y)** based on:

- X_1 = Advertising Cost (in ₹ thousands)
- X_2 = Number of Workers

Find the regression equation and predict the profit for advertising cost 20 thousands with 25 workers.

Observation	X_1 (Ad Cost)	X_2 (Workers)	Y (Profit)
1	2	5	20
2	4	6	28
3	6	7	35
4	8	9	45
5	10	10	50

Step-1: Regression Model

$$Y = b_0 + b_1X_1 + b_2X_2$$

Where:

- $b_0 \rightarrow$ Intercept
- $b_1 \rightarrow$ Coefficient of Advertising Cost
- $b_2 \rightarrow$ Coefficient of Workers

Step-2: Construct Normal Equations

Equation 1 (Overall balance):

$$\sum Y = nb_0 + b_1 \sum X_1 + b_2 \sum X_2$$

Equation 2 (Relation with X_1):

$$\sum X_1 Y = b_0 \sum X_1 + b_1 \sum X_1^2 + b_2 \sum X_1 X_2$$

Equation 3 (Relation with X_2):

$$\sum X_2 Y = b_0 \sum X_2 + b_1 \sum X_1 X_2 + b_2 \sum X_2^2$$

Step-3: Create Calculation Table

X_1	X_2	Y	X_1^2	X_2^2	X_1X_2	X_1Y	X_2Y
2	5	20	4	25	10	40	100
4	6	28	16	36	24	112	168
6	7	35	36	49	42	210	245
8	9	45	64	81	72	360	405
10	10	50	100	100	100	500	500
$\Sigma = 30$	$\Sigma = 37$	$\Sigma = 178$	$\Sigma = 220$	$\Sigma = 291$	$\Sigma = 248$	$\Sigma = 1222$	$\Sigma = 1418$

Step-4: Substitute into Normal Equations

- $178 = 5b_0 + 30b_1 + 37b_2$
- $1222 = 30b_0 + 220b_1 + 248b_2$
- $1418 = 37b_0 + 248b_1 + 291b_2$

Step-5: Solve the Equations

Equation (1):

$$5b_0 + 30b_1 + 37b_2 = 178$$

Equation (2):

$$30b_0 + 220b_1 + 248b_2 = 1222$$

Equation (3):

$$37b_0 + 248b_1 + 291b_2 = 1418$$

Solving step-by-step gives:

$$b_0 = 4.32, b_1 = 2.34, b_2 = 2.33$$

Final Regression Equation

$$Y = 4.32 + 2.34X_1 + 2.33X_2$$

$$Profit = 4.32 + 2.34(20) + 2.33(25) = 109370$$

Use Cases

1. **Real Estate:** Predict property prices using location, size and other factors.
2. **Finance:** Forecast stock prices using interest rates and inflation data.
3. **Agriculture:** Estimate crop yield from rainfall, temperature and soil quality.
4. **E-commerce:** Analyse how price, promotions and seasons affect sales.

Once you understand linear regression and its types, the next step is building the model in practice.

Cost function for Linear Regression

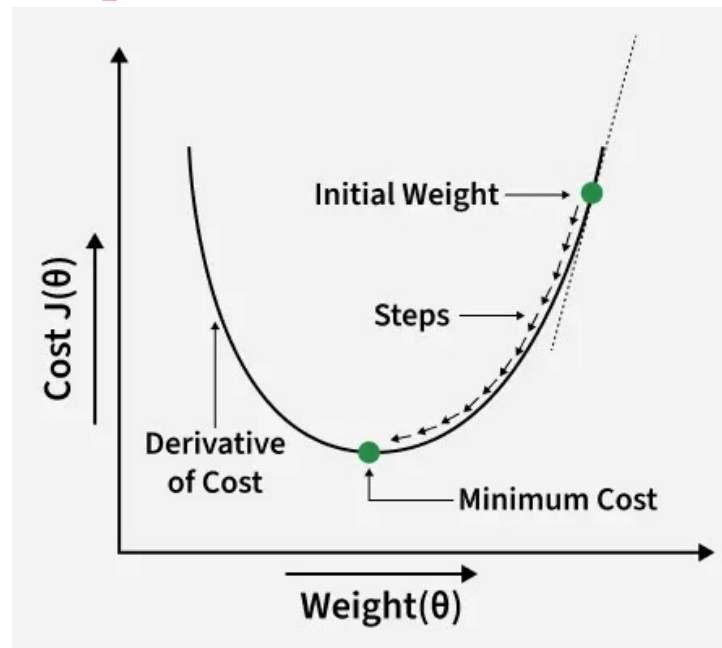
- In Linear Regression, the cost function measures how far the predicted values ($\hat{y}$) are from the actual values (y).
- It helps identify and reduce errors to find the best-fit line.
- The most common cost function used is Mean Squared Error (MSE), which calculates the average of squared differences between actual and predicted values:

$$\text{Cost function } (J) = \frac{1}{n} \sum_n^i (\hat{y}_i - y_i)^2$$

- Here, $\hat{y}_i = \theta_1 + \theta_2 x_i$
- To minimize this cost, we use Gradient Descent, which iteratively updates θ_1 and θ_2 until the MSE reaches its lowest value.
- This ensures the line fits the data as accurately as possible.

Gradient Descent for Linear Regression

- Gradient descent is an optimization technique used to train a linear regression model by minimizing the prediction error.
- It works by starting with random model parameters and repeatedly adjusting them to reduce the difference between predicted and actual values.



Working of Gradient Descent

1. Start with random values for slope and intercept.
2. Calculate the error between predicted and actual values.
3. Find how much each parameter contributes to the error (gradient).
4. Update the parameters in the direction that reduces the error.
5. Repeat until the error is as small as possible.
6. This helps the model find the best-fit line for the data.

Example of Gradient Descent for Linear Regression

- We want to predict house price based on size using a linear model:

$$\hat{y} = \theta_0 + \theta_1 x$$

- Where

- x = house size
- y = house price
- θ_0 = bias (intercept)
- θ_1 = slope

x (Size in square feet)	y (Price in lakhs)
1000	50
1500	75
2000	100

Example of Gradient Descent for Linear Regression

- **Cost Function (Mean Squared Error)**

$$J(\theta_0, \theta_1) = \frac{1}{2n} \sum_{i=1}^n (\hat{y}_i - y_i)^2$$

- Here, $n = 3$

- **Gradient Descent Update Rules**

$$\theta_0 = \theta_0 - \alpha \frac{\partial J}{\partial \theta_0}$$

$$\theta_1 = \theta_1 - \alpha \frac{\partial J}{\partial \theta_1}$$

- Where learning rate $\alpha = 0.1$ (Step Size).

Example of Gradient Descent for Linear Regression

- Partial derivatives:

$$\frac{\partial J}{\partial \theta_0} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)$$

$$\frac{\partial J}{\partial \theta_1} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i) x_i$$

- So, the Gradient Descent update rule becomes:

$$\theta_0 = \theta_0 - \alpha \left(\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i) \right)$$

$$\theta_1 = \theta_1 - \alpha \left(\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i) x_i \right)$$

Example of Gradient Descent for Linear Regression

First Iteration

1. Initialize Parameters:

$$\theta_0 = 0, \theta_1 = 0$$

Now, after putting $\theta_0 = 0$ and $\theta_1 = 0$ in regression equation,

$$\hat{y} = 0$$

2. Errors:

- $(0 - 50) = -50$
- $(0 - 75) = -75$
- $(0 - 100) = -100$

Example of Gradient Descent for Linear Regression

3. Gradients:

$$\frac{\partial J}{\partial \theta_0} = \frac{1}{3}(-50 - 75 - 100) = -75$$

$$\frac{\partial J}{\partial \theta_1} = \frac{1}{3}((-50 * 1000) + (-75 * 1500) + (-100 * 2000))$$

$$\frac{\partial J}{\partial \theta_1} = \frac{1}{3}(-50000 - 112500 - 200000) = -120833.33$$

4. Update Parameters:

$$\theta_0 = 0 - 0.1(-75) = 7.5$$

$$\theta_1 = 0 - 0.1(-120833.33) = 12083.333$$

Example of Gradient Descent for Linear Regression

5. **New Model After One Step:**

$$\hat{y} = 7.5 + 12083.333x$$

6. **After many iterations, Gradient Descent will converge to:**

$$\hat{y} \approx 0.05x$$

- It matches the true relationship.
- Gradient Descent iteratively updates parameters to minimize prediction error.

Important Note

Expression	Usage
$\frac{1}{n} \sum (error)^2$	Pure MSE
$\frac{1}{2n} \sum (error)^2$	Cost function for Gradient Descent

Question

A real estate company wants to predict house prices based on the size of the house. The following dataset is collected:

Size of House (in 1000 Sq. Ft)	Price (in lakhs)
1.0	50
1.5	75
2.0	100

Assume initial values $\theta_0 = 0$, $\theta_1 = 0$ and learning rate $\alpha = 0.1$. Perform **two iteration** of gradient descent and calculate the updated values of θ_0 and θ_1 .

Solution

After Iteration 1:

$$\theta_0 = 7.5, \theta_1 = 12.08$$

New Hypothesis:

$$\hat{y} = 7.5 + 12.08x$$

After Iteration 2:

$$\theta_0 = 12.44, \theta_1 = 20.12$$

New Hypothesis:

$$\hat{y} = 12.44 + 20.12x$$

Evaluation Metrics for Linear Regression

- A variety of evaluation measures can be used to determine the strength of any linear regression model.
- These assessment metrics often give an indication of how well the model is producing the observed outputs.
- The most common measurements are:

1. Mean Square Error (MSE)

- MSE is an evaluation metric that calculates the **average of the squared differences between the actual and predicted values for all the data points.**
- The difference is **squared to ensure that negative and positive differences don't cancel each other out.**

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- Here,
 - n is the number of data points.
 - y_i is the actual or observed value for the i^{th} data point.
 - $\hat{y}_i$ is the predicted value for the i^{th} data point.
- MSE is a way to **quantify the accuracy** of a model's predictions.
- MSE is **sensitive to outliers** as large errors contribute significantly to the overall score.

2. Mean Absolute Error (MAE)

- MAE is an evaluation metric used to calculate the accuracy of a regression model.
- MAE measures the average absolute difference between the predicted values and actual values.
- Mathematically MAE is expressed as:

$$MAE = \frac{1}{n} \sum_{i=1}^n |Y_i - \hat{Y}_i|$$

- Here,
 - n is the number of observations
 - Y_i represents the actual values.
 - $\hat{Y}_i$ represents the predicted values
- Lower MAE value indicates better model performance.
- It is not sensitive to the outliers as we consider absolute differences.

3. Root Mean Squared Error (RMSE)

- The **square root of the residuals' variance** is the RMSE.
- It describes **how well the observed data points match the expected values or the model's absolute fit to the data.**
- In mathematical notation, it can be expressed as:

$$RMSE = \sqrt{MSE} = \sqrt{\frac{\sum_{i=1}^n (y_i^{actual} - y_i^{predicted})^2}{n}}$$

- Where:
 - n is the Number of observations
 - y_i^{actual} is the Actual value
 - $y_i^{predicted}$ is the Predicted value

4. Coefficient of Determination (R-squared)

- R-Squared is a statistic that indicates how much variation the developed model can explain or capture.
- It is always in the range of 0 to 1.
- In general, the better the model matches the data, the greater the R-squared number.
- In mathematical notation, it can be expressed as:

$$R^2 = 1 - \left(\frac{RSS}{TSS} \right)$$

4. Coefficient of Determination (R-squared)

RSS:

- The sum of squares of the residual for each data point in the plot or data is known as the residual sum of squares or RSS.
- It is a measurement of the difference between the output that was observed and what was anticipated.

$$RSS = \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2$$

4. Coefficient of Determination (R-squared)

TSS:

- The sum of the data points' errors from the output variable's mean is known as the total sum of squares or TSS.

$$TSS = \sum_{i=1}^n (y_i - \bar{y})^2$$

- It measures the total variation present in the target variable y .
- R squared metric is a measure of the proportion of variance in the dependent variable that is explained the independent variables in the model.

5. Adjusted R-Squared Error

- Adjusted R^2 measures the proportion of variance in the dependent variable that is explained by independent variables in a regression model.
- It accounts the number of predictors in the model and penalizes the model for including irrelevant predictors that don't contribute significantly to explain the variance in the dependent variables.
- Mathematically, adjusted R^2 is expressed as:

$$\text{Adjusted } R^2 = 1 - \left(\frac{(1 - R^2) \cdot (n - 1)}{n - k - 1} \right)$$

- Here,
 - n is the number of observations.
 - k is the number of predictors in the model.
 - R^2 is coefficient of determination.
- It penalizes the inclusion of unnecessary predictors, helping to prevent overfitting.

Important Note

Case	Result
Adjusted R^2 increases	New predictor is useful
Adjusted R^2 decreases	New predictor is irrelevant
Adjusted $R^2 \approx R^2$	Model is well balanced

What is Underfitting?

- Underfitting happens when the model fails to learn important patterns.
- It performs poorly on both training and testing data.
- Underfitting happens due to:
 - Very high regularization
 - Features are weak or missing
 - Not enough training
 - High bias

Important Terms

- **Bias:** It is like assuming all birds can only be small and fly, so the model fails to recognize big birds like ostriches or penguins that can't fly and get biased with predictions.
- **Variance:** Error that happens when a machine learning model learns too much from the training data, including random noise.

Bias–Variance Inside Underfitting

- Underfitting mainly occurs due to:
 - High bias means model makes strong assumptions
 - Ignores patterns
 - Learns an overly simple representation
 - Variance is low because the model gives similar outputs even if the data changes

Underfitting = High Bias + Low Variance

What is Overfitting?

- Overfitting happens when the model **learns too much from the training data**, including noise and outliers.
- It performs very well on training data but poorly on test data.
- Overfitting happens due to:
 - Model too complex
 - Too many features
 - Very little data
 - No regularization
 - **High variance**

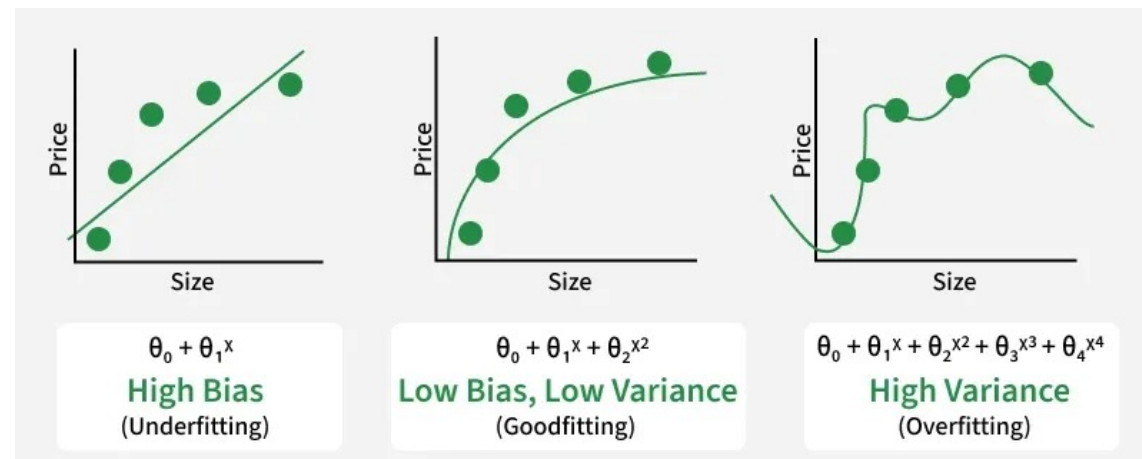
Bias–Variance Inside Overfitting

- Overfitting is mainly caused by:
 - High variance means **model reacts too strongly to training data**
 - Learns noise as patterns
 - **Low bias because the model is extremely flexible**

$$\textit{Overfitting} = \textit{Low Bias} + \textit{High Variance}$$

Underfitting v/s Overfitting v/s Appropriate Fitting

- **Underfitting:** Straight line trying to fit a curved dataset but cannot capture the data's patterns, leading to poor performance on both training and test sets.
- **Overfitting:** A squiggly curve passing through all training points, failing to generalize performing well on training data but poorly on test data.
- **Appropriate Fitting:** Curve that follows the data trend without overcomplicating to capture the true patterns in the data.



Bias-Variance Trade-off

- The relationship between bias and variance is often referred to as the bias-variance tradeoff, which highlights the need for balance:
 - Increasing model complexity reduces bias but increases variance (risk of overfitting).
 - Simplifying the model reduces variance but increases bias (risk of underfitting).
- The goal is to find an optimal balance where both bias and variance are minimized, resulting in good generalization performance.

Summary Table

Model Type	Bias	Variance	Training Error	Testing Error
Underfitting	High	Low	High	High
Overfitting	Low	High	Low	High
Good Fit	Low	Low	Low	Low

Techniques to Reduce Underfitting

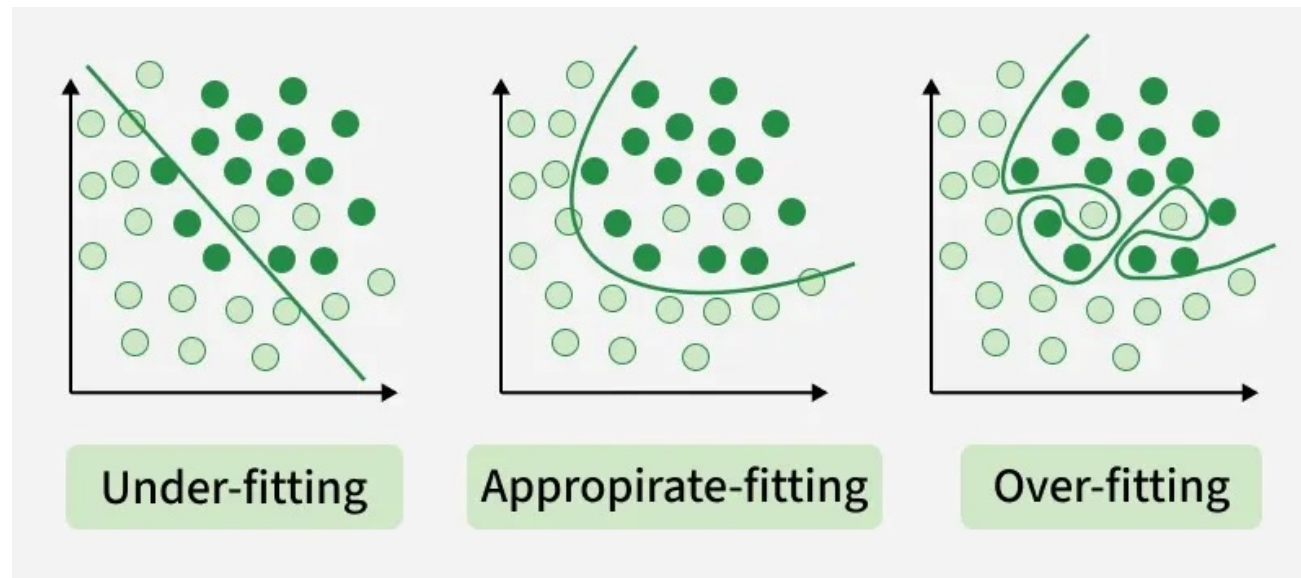
- Use a more complex model
- Add new features and perform feature engineering
- Reduce regularization
- Train for more epochs

Techniques to Reduce Overfitting

- Collect more training data
- Reduce model complexity
- Use regularization (L1 / L2)
- Use early stopping
- Clean noisy data

Regularization in Machine Learning

- Regularization is a technique used in machine learning to prevent overfitting, which otherwise causes models to perform poorly on unseen data.
- By adding a penalty for complexity, regularization encourages simpler and more generalizable models.



Regularization Techniques for Linear Models

1. Lasso Regression (L1 Regularization)

- LASSO (Least Absolute Shrinkage and Selection Operator) Regression is a technique used for regularizing a linear regression model.
- This penalty can shrink some coefficients to zero which helps in selecting only the important features and ignoring the less important ones.
- The objective function after applying lasso regression is:

$$J(\theta) = \frac{1}{2n} \sum_{i=1}^n (\hat{y}_i - y_i)^2 + \frac{\lambda}{n} \sum_{j=1}^m |\theta_j|$$

- The first term is the least squares loss, representing the squared difference between predicted and actual values.
- The second term is the L1 regularization term, it penalizes the sum of absolute values of the regression coefficient θ_j .

Regularization Techniques for Linear Models

2. Ridge Regression (L2 Regularization)

- Ridge regression is a linear regression technique that adds a regularization term to the standard linear objective.
- Again, the goal is to prevent overfitting by penalizing large coefficient in linear regression equation.
- It handles multicollinearity by shrinking the coefficients of correlated features instead of eliminating them.
- The objective function after applying ridge regression is:

$$J(\theta) = \frac{1}{2n} \sum_{i=1}^n (\hat{y}_i - y_i)^2 + \frac{\lambda}{2n} \sum_{j=1}^m \theta_j^2$$

- The first term is the least squares loss, representing the squared difference between predicted and actual values.
- The second term is the L2 regularization term, it penalizes the sum of square of values of the regression coefficient θ_j .

Regularization Techniques for Linear Models

3. Elastic Net Regression

- Elastic Net Regression is a **hybrid regularization technique** that combines the power of both L1 and L2 regularization in linear regression objective.

$$J(\theta) = \frac{1}{2n} \sum_{i=1}^n (\hat{y}_i - y_i)^2 + \alpha \left(\frac{\lambda}{n} \sum_{j=1}^m |\theta_j| \right) + (1 - \alpha) \left(\frac{\lambda}{2n} \sum_{j=1}^m \theta_j^2 \right)$$

- The first term is least square loss.
- The second term is L1 regularization and third is ridge regression.
- λ is the overall regularization strength.**
- α controls the mix between L1 and L2 regularization.

Benefits of Regularization

1. **Prevents Overfitting:** Regularization helps models focus on underlying patterns instead of memorizing noise in the training data.
2. **Enhances Performance:** Prevents excessive weighting of outliers or irrelevant features helps in improving overall model accuracy.
3. **Stabilizes Models:** Reduces sensitivity to minor data changes which ensures consistency across different data subsets.

Benefits of Regularization

4. **Prevents Complexity:** Keeps model from becoming too complex which is important for limited or noisy data.
5. **Handles Multicollinearity:** Reduces the magnitudes of correlated coefficients helps in improving model stability.
6. **Promotes Consistency:** Ensures reliable performance across different datasets which reduces the risk of large performance shifts.

Lasso vs Ridge vs Elastic Net

Features	Lasso Regression	Ridge Regression	Elastic Net Regression
Penalty Type	L1 Penalty: Lasso uses the absolute values of coefficients.	L2 Penalty: Ridge uses the square of the coefficients.	L1 + L2 Penalty: Elastic Net uses both absolute and square penalties together.
Effect on Coefficients	It completely removes unnecessary features by setting their coefficients to zero.	It makes all coefficients smaller but doesn't set them to zero.	It removes some features and reduces others by balancing both.
Best Use Case	It is best when we want to remove irrelevant features	It is good when all features matter but we want to reduce their impact	It is best for when we features are correlated and feature selection is needed

Lasso vs Ridge vs Elastic Net

Features	Lasso Regression	Ridge Regression	Elastic Net Regression
Hyperparameters involved	Delta: Controls how much regularization is applied. A higher alpha means more shrinkage.	Delta: Similar to Lasso which helps in controlling the strength of regularization.	Delta + Alpha: Two parameters. Delta controls regularization strength and Alpha adjusts the balance between Lasso and Ridge.
Strengths	It is great for automatically choosing important features.	It works well when features are related but shouldn't be completely removed.	It combines Lasso's feature selection and Ridge's handling of correlations.

Lasso vs Ridge vs Elastic Net

Features	Lasso Regression	Ridge Regression	Elastic Net Regression
Weaknesses	It can sometimes remove useful features if not tuned properly.	It keeps all features which may not help in high-dimensional data with irrelevant features.	It is a bit harder to tune due to having two parameters.
Example	Imagine we have 100 features to predict house prices. It will set the coefficients of irrelevant features (like house color) to zero.	If we have 100 features it will reduce the impact of every feature but won't completely remove any.	If we have features like "size" and "rooms" that are similar it will remove one and shrink the other.

Using the right regularization technique helps us to build models that are both accurate and easy to interpret.

Numerical on Regularization in Machine Learning

Important Note:

- λ is divided by $2m$ to normalize the regularization term with respect to the number of training samples and to simplify gradient calculations by cancelling constants during differentiation.
- In regularized linear regression, if cost part is averaged then loss function part should be averaged.
 - In case of L1 penalty, $\frac{\lambda}{n}$
 - In L2 penalty, $\frac{\lambda}{2n}$
 - Here, n is total number of samples

Question 1

For a linear regression model with $m = 5$ training examples, the learned parameters are:

$$\theta_0 = 1, \theta_1 = 2, \theta_2 = -1$$

The regularization parameter is $\lambda = 10$. Calculate the L2 regularization term added to the cost function.

Solution 1

- L2 regularization penalty is:
- $\frac{\lambda}{2m}(\theta_1^2 + \theta_2^2)$
- $\theta_1^2 + \theta_2^2 = 2^2 + (-1)^2 = 4 + 1 = 5$
- $\frac{\lambda}{2m} = \frac{10}{2 \times 5} = 1$
- L2 penalty = $1 \times 5 = 5$

Question 2

Given $m = 4$, $\lambda = 8$, and model parameters:

$$\theta_0 = 0.5, \theta_1 = -2, \theta_2 = 1, \theta_3 = 0$$

Compute the L1 regularization term.

Solution 2

- L1 regularization penalty:
- $\frac{\lambda}{m} (|\theta_1| + |\theta_2| + |\theta_3|)$
- $|\theta_1| + |\theta_2| + |\theta_3| = 2 + 1 + 0 = 3$
- $\frac{\lambda}{m} = \frac{8}{4} = 2$
- L1 penalty = $2 \times 3 = 6$

Question 3

For the parameter vector:

$$\theta = [3, -1, 0, 2]$$

Calculate:

- L1 penalty
- L2 penalty

Solution 3

- L1 penalty

$$|3| + |-1| + |0| + |2| = 3 + 1 + 0 + 2 = 6$$

- L2 penalty

$$3^2 + (-1)^2 + 0^2 + 2^2 = 9 + 1 + 0 + 4 = 14$$

Question 4

Consider a linear regression model with $m = 5$ training examples. The learned parameters are:

$$\theta_0 = 1, \theta_1 = 2, \theta_2 = -1, \theta_3 = 0$$

Elastic Net regularization is used with:

- Regularization parameter $\lambda = 6$
- Mixing parameter $\alpha = 0.4$

Calculate the Elastic Net regularization penalty added to the cost function.

Solution 4

1. $\text{Penalty} = \left[\alpha * \frac{\lambda}{m} \sum |\theta_j| + (1 - \alpha) * \frac{\lambda}{2m} \sum \theta_j^2 \right]$
2. Compute L1 part
3. $\sum |\theta_j| = |2| + |-1| + |0| = 2 + 1 + 0 = 3$
4. $\alpha \sum |\theta_j| = 0.4 \times 3 = 1.2$
5. $\frac{\lambda}{m} = \frac{6}{5} = 1.2$
6. L1 penalty = $1.2 * 1.2 = 1.44$
7. Compute L2 part
8. $\sum \theta_j^2 = 2^2 + (-1)^2 + 0^2 = 4 + 1 + 0 = 5$
9. $(1 - \alpha) \sum \theta_j^2 = 0.6 \times 5 = 3$
10. $\frac{\lambda}{2m} = \frac{6}{10} = 0.6$
11. L2 penalty = $0.6 * 3 = 1.8$
12. Combine L1 and L2 terms
13. Elastic Net penalty = $1.44 + 1.8 = 3.24$

MCQ 1

In Elastic Net regularization, when the mixing parameter $\alpha = 1$, the method becomes:

- A. Ridge regression
- B. Elastic Net
- C. Lasso regression
- D. No regularization

MCQ 2

If $\alpha = 0$ in Elastic Net, which penalty term remains active?

- A. Only L1 penalty
- B. Only L2 penalty
- C. Both L1 and L2 penalties
- D. No penalty

MCQ 3

Which regularization technique is most suitable when predictors are highly correlated and feature selection is required?

- A. Ridge
- B. Lasso
- C. Elastic Net
- D. Ordinary Least Squares

MCQ 4

Which of the following parameters is NOT regularized in Elastic Net regression?

- A. θ_0
- B. θ_1
- C. θ_2
- D. θ_n

Why Linear Regression is Important

1. **Simplicity and Interpretability:** It's **easy to understand and interpret**, making it a starting point for learning about machine learning.
2. **Predictive Ability:** Helps **predict future outcomes based on past data**, making it useful in various fields like finance, healthcare and marketing.
3. **Basis for Other Models:** Many advanced algorithms, like logistic regression or neural networks, **build on the concepts of linear regression**.

Why Linear Regression is Important

4. **Efficiency:** It's **computationally efficient** and works well for problems with a linear relationship.
5. **Widely Used:** It's one of the most **widely used techniques in both statistics and machine learning** for regression tasks.
6. **Analysis:** It provides insights into **relationships between variables** (e.g., how much one variable influences another).

Advantages

1. Linear regression is a relatively **simple algorithm**, making it **easy to understand and implement**.
2. The coefficients of the linear regression model can be interpreted as the change in the dependent variable for a one-unit change in the independent variable, **providing insights into the relationships between variables**.
3. Linear regression is **computationally efficient** and can handle large datasets effectively. It can be trained quickly on large datasets, making it **suitable for real-time applications**.
4. Linear regression is relatively **robust to outliers** compared to other machine learning algorithms. Outliers may have a smaller impact on the overall model performance.

Advantages

4. Linear regression often serves as a **good baseline model for comparison** with more complex machine learning algorithms.
5. Linear regression is a well-established algorithm with a rich history and is widely available in various machine learning libraries and software packages.

Limitations

1. Linear regression **assumes a linear relationship** between the dependent and independent variables. If the relationship is not linear, the model may not perform well.
2. Linear regression is **sensitive to multicollinearity**, which occurs when there is a high correlation between independent variables. Multicollinearity can inflate the variance of the coefficients and lead to unstable model predictions.
3. Linear regression assumes that the features are already in a suitable form for the model. **Feature engineering may be required** to transform features into a format that can be effectively used by the model.

Limitations

4. Linear regression is susceptible to both overfitting and underfitting. Overfitting occurs when the model learns the training data too well and fails to generalize to unseen data. Underfitting occurs when the model is too simple to capture the underlying relationships in the data.
5. Linear regression provides limited explanatory power for complex relationships between variables. More advanced machine learning techniques may be necessary for deeper insights.

Thank You!